

CHERKASY, Ukraine

WMO: 334870

Lat: 49.4092N

Lon: 32.0031E

Elev: 107

StdP: 100.05

Time Zone: 2.00 (EUE)

Period: 84-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-19.3	-15.9	-22.0	0.5	-18.9	-18.9	0.7	-15.0	15.3	-5.2	11.8	-1.4	2.7	300	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	30.4	19.5	28.8	18.8	27.4	18.2	21.1	27.1	20.2	26.2	19.4	25.2	3.7	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.2	14.2	23.5	18.1	13.2	22.4	17.2	12.5	21.9	61.6	27.2	58.3	26.2	55.7	25.4	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.8	8.3	7.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.5	-3.3	-3.0	1.9	9.3	15.8	18.1	19.9	19.6	14.6	8.3	2.3	-1.9
	DBStd	9.73	5.91	5.96	4.94	4.07	3.67	3.06	3.10	3.35	3.77	4.59	5.31	4.65
	HDD10.0	1791	413	364	253	62	3	0	0	0	6	88	233	368
	HDD18.3	3776	671	598	510	273	97	41	19	26	122	313	481	627
	CDD10.0	1251	0	0	2	40	182	244	308	296	144	33	2	0
	CDD18.3	196	0	0	0	1	18	34	68	63	10	1	0	0
	CDH23.3	1928	0	0	0	16	143	283	651	720	114	1	0	0
	CDH26.7	505	0	0	0	2	16	58	168	238	23	0	0	0
(14) Wind	WSAvg	3.6	4.2	3.8	4.2	3.9	3.4	3.3	3.2	2.9	3.3	3.5	3.6	3.9
(15) Precipitation	PrecAvg	567	40	36	33	41	48	67	73	54	43	32	43	48
	PrecMax	794	160	86	65	99	114	151	147	120	179	119	85	98
	PrecMin	357	8	0	1	9	4	0	12	6	4	0	6	7
	PrecStd	104	28	22	18	21	27	31	35	30	39	27	21	21
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.9	9.9	16.9	25.8	28.3	30.3	31.5	33.2	29.1	22.4	15.2	9.1
		MCWB	5.6	7.3	9.6	16.0	17.0	19.4	20.2	19.2	18.0	15.2	11.6	6.5
	2%	DB	5.0	7.1	13.9	21.0	26.7	28.2	29.9	31.1	26.4	19.8	12.9	7.1
		MCWB	3.7	5.7	8.5	12.5	16.6	18.9	19.7	19.3	17.0	14.7	10.6	5.3
	5%	DB	3.8	5.1	10.9	18.8	24.9	26.5	28.3	29.2	24.2	17.5	10.8	5.0
		MCWB	2.7	3.8	6.8	11.6	16.8	18.1	19.1	18.8	16.8	13.3	8.7	3.8
	10%	DB	2.8	3.4	8.3	16.5	22.9	24.7	26.9	27.3	21.8	15.1	8.8	3.4
		MCWB	1.8	2.1	5.4	10.3	15.9	17.7	18.8	17.9	15.4	12.3	7.3	2.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.0	7.7	10.8	15.5	19.4	21.6	23.2	21.8	19.8	17.3	12.7	7.2
		MCDB	7.5	9.0	16.1	24.0	24.1	25.6	28.5	27.8	26.5	20.1	14.8	8.7
	2%	WB	3.8	6.1	8.9	13.8	18.4	20.2	21.5	20.6	18.3	15.3	11.0	5.5
		MCDB	4.5	7.3	12.9	19.2	23.7	25.7	27.1	27.6	23.9	18.2	12.7	6.9
	5%	WB	2.9	3.9	7.2	12.3	17.5	19.3	20.5	19.7	17.1	13.9	8.9	3.9
		MCDB	3.6	5.1	10.3	17.3	23.0	24.6	26.1	26.6	22.3	16.7	10.5	4.9
	10%	WB	1.9	2.2	5.7	11.2	16.5	18.4	19.5	18.8	16.1	12.5	7.3	2.5
		MCDB	2.6	3.1	8.2	16.1	21.6	23.2	24.9	25.3	20.9	14.9	8.6	3.3
(35) Mean Daily Temperature Range	5% DB	MDBR	4.9	6.0	7.2	9.5	10.6	9.6	11.0	12.0	10.4	9.2	5.6	4.4
		MCDBR	4.6	6.7	11.8	13.8	12.2	12.6	13.1	15.0	14.3	13.2	8.8	6.2
		MCWBR	3.8	5.2	7.1	7.0	4.9	5.4	4.9	5.4	6.6	8.1	6.6	5.0
	5% WB	MCDBR	4.6	6.7	10.6	12.3	10.7	10.5	11.4	13.1	13.0	10.5	7.2	5.7
		MCWBR	4.0	5.5	6.9	6.7	5.0	5.3	5.2	5.4	6.8	7.1	5.6	4.7
(40) Clear-Sky Solar Irradiance	taub		0.319	0.352	0.384	0.446	0.414	0.407	0.436	0.436	0.404	0.369	0.350	0.321
	taud		2.194	2.151	2.197	2.096	2.275	2.315	2.243	2.229	2.281	2.353	2.304	2.260
	Ebn at Noon		700	767	813	797	847	854	820	800	786	750	666	646
	Edh at Noon		78	105	119	146	128	123	131	127	109	86	72	65
(44) All-Sky Solar Radiation	RadAvg		0.94	1.74	2.98	4.45	5.79	6.17	6.01	5.36	3.61	2.13	0.88	0.65
	RadStd		0.18	0.28	0.46	0.42	0.47	0.49	0.36	0.33	0.53	0.31	0.14	0.16

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)	+0.72	N/A	N/A	N/A	+1.02	N/A	N/A	-133	-185	+118	+68

Nomenclature: See separate page